

Jackie Lok

CURRICULUM VITAE

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EMPLOYMENT

National University of Singapore, Department of Mathematics, *Singapore*
Peng Tsu Ann Assistant Professor (Visiting Fellow) 2026–

EDUCATION

Princeton University, *Princeton, NJ, USA*
Ph.D. in Operations Research and Financial Engineering 2021–2026
Adviser: [Elizaveta Rebrova](#)

M.A. in Operations Research and Financial Engineering 2023

UNSW Sydney, *Sydney, Australia*
Bachelor of Science (Honours) in Pure Mathematics 2020
with First Class Honours and the University Medal
Supervisor: [Catherine Greenhill](#)
WAM: 97.16/100

Bachelor of Actuarial Studies (Co-op) in Mathematics with Distinction 2016–2019
WAM: 95.65/100

Wharton School, University of Pennsylvania, *Philadelphia, PA, USA*
International Exchange Semester, GPA: 4.00/4.00 2017

RESEARCH INTERESTS

My research interests broadly lie at the intersection of probability, statistics, and optimization. I am interested in analyzing randomized algorithms in numerical linear algebra, machine learning, and data science, with a focus on developing theory and tools that allow us to better understand and work with large-scale or high-dimensional data.

AWARDS & HONOURS

- Quad Fellowship, Schmidt Futures 2023
- Richard Stillwell '21 *24 and Agnes Newhall Stillwell Fellowship, Princeton University 2021
- University Medal in Pure Mathematics, UNSW Sydney 2020
- H.C. & M.E. Porter Memorial Scholarship, UNSW Sydney 2020
- The Faculty of Science Prize for Honours Year Science, UNSW Sydney 2020
- The George Szekeres Prize, UNSW Sydney 2019
- The Head of School's Prize, UNSW Sydney 2019
- UNSW Co-op Scholarship in Actuarial Studies 2016–2019
- UNSW Scientia Scholarship 2016–2019
- Harry Manson International Exchange Scholarship, UNSW Sydney 2017

Travel grants/awards

SEAS Travel Grant, Princeton University (2024, 2025). SIAM Student Travel Award (2024). ANU Annual Graduate School in Mathematical Aspects of Data Science Participant Support (2024)

PUBLICATIONS

Journal papers

- Jackie Lok and Elizaveta Rebrova. “Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations”. *SIAM Journal on Matrix Analysis and Applications* 47.3 (2026), pp. 1495–1529. DOI: [10.1137/25M177446X](https://doi.org/10.1137/25M177446X). arXiv: [2506.09394](https://arxiv.org/abs/2506.09394).
- Roxanne He and Jackie Lok. “On approximating the Potts model with contracting Glauber dynamics”. *Advances in Applied Probability* 58.2 (2026), pp. 812–849. DOI: [10.1017/apr.2025.10036](https://doi.org/10.1017/apr.2025.10036). arXiv: [2404.18778](https://arxiv.org/abs/2404.18778).
- Jackie Lok and Elizaveta Rebrova. “A subspace constrained randomized Kaczmarz method for structure or external knowledge exploitation”. *Linear Algebra and its Applications* 698 (2024), pp. 220–260. DOI: [10.1016/j.laa.2024.06.010](https://doi.org/10.1016/j.laa.2024.06.010). arXiv: [2309.04889](https://arxiv.org/abs/2309.04889).

Conference papers

- Jackie Lok, Rishi Sonthalia, and Elizaveta Rebrova. “Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression”. *Proceedings of the 36th International Conference on Algorithmic Learning Theory (ALT 2025)*. Vol. 272. PMLR, 2025. arXiv: [2406.03696](https://arxiv.org/abs/2406.03696).

Preprints

- Toby Anderson, Max Collins, Jamie Haddock, Jackie Lok, and Elizaveta Rebrova. “Beyond Expectation: Concentration Inequalities for Randomized Iterative Methods” (2025). arXiv: [2511.20877](https://arxiv.org/abs/2511.20877).
- Rishi Sonthalia, Jackie Lok, and Elizaveta Rebrova. “On Regularization via Early Stopping for Least Squares Regression” (2024). arXiv: [2406.04425](https://arxiv.org/abs/2406.04425).

Miscellaneous

- Jackie Lok. *Randomized Iterative Algorithms in Numerical Linear Algebra*. PhD thesis. 2026.
- Jackie Lok. *Markov chains, mixing times, and cutoff*. Honours thesis. 2020.

TALKS AND PRESENTATIONS

- AMS Postdoc Seminar, Johns Hopkins University: “Subspace-constrained sketch-and-project solvers for linear systems with approximately low-rank structure or external knowledge” (virtual) Apr 2026
- ICERM Workshop on Randomized Numerical Linear Algebra, Providence, RI: “Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations” (poster) Feb 2026
- Joint Mathematics Meetings (JMM 2026), Washington DC: “Efficient linear solvers for approximately low-rank linear systems” Jan 2026
- Mid-Atlantic Numerical Analysis Day, Temple University: “Sketch-and-Project Solvers for Linear Systems with Low-Rank Structure” Nov 2025
- Simons Institute Workshop on Linear Systems and Eigenvalue Problems, UC Berkeley: “Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations” (poster) Oct 2025
- The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Canada: “Subspace-constrained Sketch-and-project Solvers for Linear Systems with Low-rank Structure” Jul 2025
- NLA group meeting, University of Oxford, UK: “Subspace-constrained sketch-and-project solvers for linear systems with low-rank structure” Jul 2025

- The 36th International Conference on Algorithmic Learning Theory (ALT 2025), Milan, Italy:
“Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression”
Feb 2025
- Conference on Mathematical Theory of Deep Neural Networks (DeepMath 2024), Philadelphia:
“Error dynamics of mini-batch gradient descent with random reshuffling for least squares” (poster)
Nov 2024
- CUNY Graduate Center Harmonic Analysis & PDE Seminar: “A subspace constrained randomized
Kaczmarz method” Nov 2024
- SIAM Conference on Mathematics of Data Science (MDS24), Atlanta: “A Subspace Constrained
Randomized Kaczmarz Method for Structure or External Knowledge Exploitation” (poster) Oct
2024
- NSF CompMath PI Meeting, University of Washington, Seattle: “A Subspace Constrained
Randomized Kaczmarz Method for Structure or External Knowledge Exploitation” (poster) Jul 2024
- Graduate student probability reading group, Princeton University:
“Concentration for Random Matrix Products” Oct 2023
“Matrix Concentration Inequalities via the Method of Exchangeable Pairs” Nov 2022
- Honours presentation, UNSW Sydney: “Mixing times of Markov chains and the cutoff phenomenon”
Nov 2020

TEACHING

National University of Singapore, *Singapore*

Lecturer, Department of Mathematics

- [MA4270: Data Modelling and Computation](#) Semester 1 2026/27

Princeton University, *Princeton, NJ, USA*

Teaching Assistant (TA), ORFE Department

Responsible for delivering weekly precepts, holding office hours, grading problem sets and exams, and general course admin.

- [ORF 405: Regression and Applied Time Series](#) Fall 2025
- [ORF 387: Networks](#) Spring 2025
- [ORF 526: Probability Theory](#) Fall 2024
- [ORF 387: Networks](#) Spring 2024
- [ORF 363: Computing and Optimization](#) Fall 2023
- [ORF 350: Analysis of Big Data](#) Spring 2023
- [ORF 387: Networks](#) Fall 2022

University of Melbourne, *Melbourne, Australia*

Academic Tutor, School of Mathematics and Statistics

Responsible for delivering weekly tutorials and marking assessments.

- MAST20004: Probability Semester 1 2021

UNSW Sydney, *Sydney, Australia*

Academic Tutor, School of Risk and Actuarial Studies

Responsible for delivering weekly tutorials, marking exams and assessments, developing course materials, holding student consultations, and providing general course support.

- ACTL3162: General Insurance Techniques Term 3 2020
- ACTL2102: Foundations of Actuarial Models Term 2 2020
- ACTL2111: Financial Mathematics for Actuaries Term 1 2020
- ACTL1101: Introduction to Actuarial Studies Term 3 2019
- ACTL2102: Foundations of Actuarial Models Term 2 2019
- ACTL3141: Actuarial Models and Statistics Term 1 2019

OTHER ACTIVITIES

Mentoring

- [McGraw Graduate Teaching Fellow](#), *Princeton University* 2024–2026
- [ReMatch mentor](#), *Princeton University* 2023
- [ORFE Senior Thesis Writer's Group](#) co-leader, *Princeton University* 2022–2023
- [Drop-in Centre tutor](#), *School of Mathematics and Statistics, UNSW Sydney* 2020

Reviewing

IMA Journal of Numerical Analysis, Journal of Computational and Applied Mathematics, Linear Algebra and its Applications, SIAM Journal on Matrix Analysis and Applications

WORK EXPERIENCE

icare, Actuarial Services Intern, *Sydney, Australia* Aug 2018–Feb 2019

Supported the provision of actuarial advice and analysis for the NSW state insurer. Assisted with the reporting and valuation of outstanding claims liabilities, scenario analysis, preparation of financial budgets, claims experience monitoring, and the assessment of data quality and integrity.

Suncorp Group, Natural Perils Pricing Intern, *Sydney, Australia* Feb 2018–Aug 2018

Collaborated in the research and development of a new natural peril pricing model in Python using analytical and machine learning techniques with insurance and geospatial datasets. Developed interactive tool using SAS and Python to identify and visualise exposure concentration risks as part of an automated monitoring pipeline.

MetLife Australia, Capital and Valuation Intern, *Sydney, Australia* Nov 2016–Feb 2017

Assisted with financial reporting, reserving and scenario analysis for group life insurance.

LANGUAGES AND SKILLS

Languages

English (native), Cantonese (fluent), Mandarin (beginner), German (beginner)

Computing

- Proficient with Python. Experience with other programming languages including Julia, R, Java, MATLAB, SQL, and SAS
- Competent with $\text{\LaTeX}$
- Experience with Microsoft Excel, Word, and PowerPoint

Online courses

- [Probabilistic Graphical Models Specialization](#) (Coursera, Stanford University) 2021
- [Deep Learning Specialization](#) (Coursera, DeepLearning.AI) 2021
- [Machine Learning](#) (Coursera, Stanford University) 2018